
DFT-Based Approaches to Improving Hydrogen Storage in Graphene-Based Materials

Presented by: Md. Nazmul Hossan Talukder Fahim (ID: ET-233053)

Source: Cruz-Martínez et al. (2024). *Molecules*, 29(2), 436.

Introduction: The Storage Challenge



The Potential

Hydrogen yields 142 MJ/kg energy density, far exceeding conventional fuels.



The Problem

Pure graphene lacks sufficient interaction to trap H₂ at ambient conditions.



The Objective

Achieve optimal adsorption energy (-0.2 to -0.6 eV) via surface modification.

Literature Review: Graphene Modification Strategies



Decoration

Depositing Transition Metals (Ti, Ca) as active centers for adsorption.



Doping

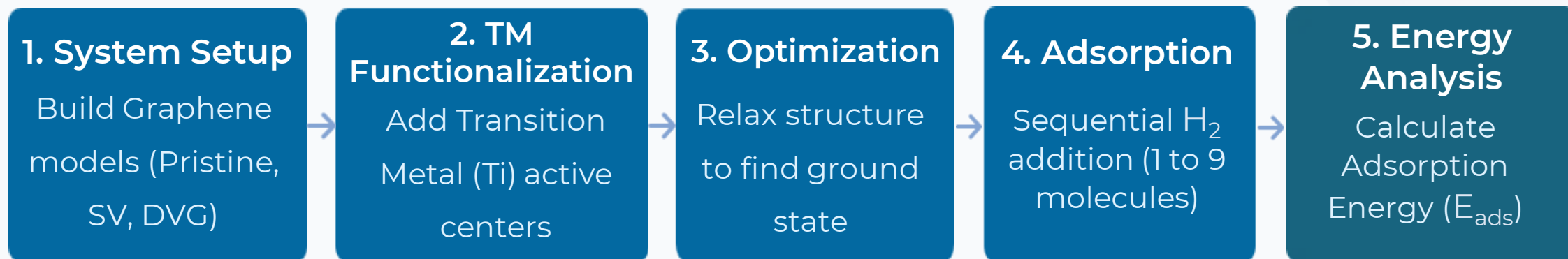
Replacing C atoms with heteroatoms (N, B) to polarize the lattice.



Defects

Creating Vacancies (SV, DVG) to trap metal atoms and prevent clustering.

Methodology: DFT Computational Flow



$$E_{\text{ads}} = E_{\text{total}} - (E_{\text{surf}} + E_{\text{mol}})$$

E_{total} = Total Ground-State Energy, E_{surf} = Energy of the surface , E_{mol} = Energy of the H₂ molecule

 Simulated using VASP with PBE functional and DFT-D3 van der Waals corrections.

Results: Adsorption Energy Analysis

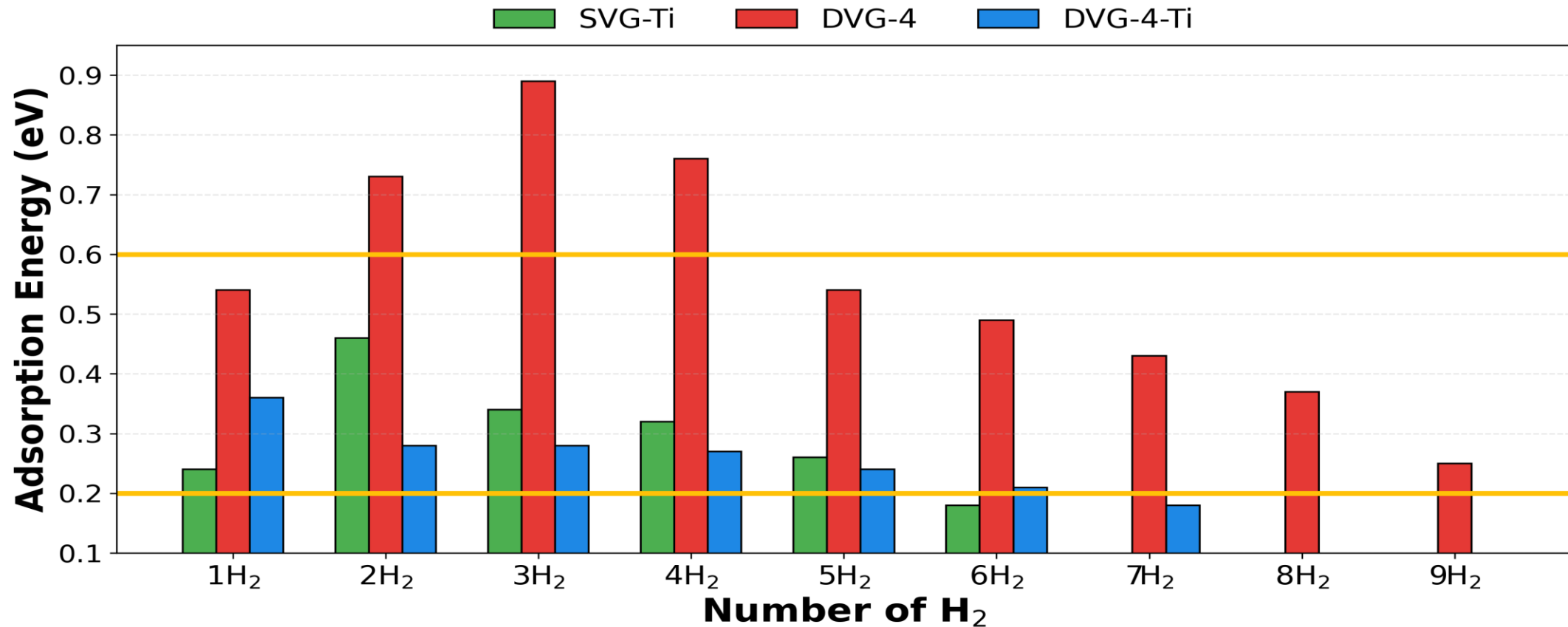


Figure-1: Hydrogen molecule adsorption energy comparison across modified graphene structures

Key Finding: **DVG-4-Ti** stays within the optimal binding range for all 9 molecules.

Conclusion

- ✓ **Stability:** Vacancy-doped structures (DVG-4-Ti) achieve a critical balance between stability and capacity.
- ✓ **Efficiency:** Binding range stays within ideal -0.2 to -0.6 eV for ambient reversible storage.
- ✓ **Capacity:** Theoretically meets DOE gravimetric targets (5.5 wt.%) for fuel cell vehicles.

Potential Application

Powering next-generation hydrogen fuel cell vehicles via stable solid-state storage.

Thank You! Questions?